

Abstract

Abstract—Unsupervised panoptic segmentation must recover both pixel-level semantics and individual object identities without human masks, and the strongest scene-centric recipe to date, CUPS, supplies the missing structure from stereo depth and scene flow at pseudo-label time — a commitment that excludes the monocular footage in which most images actually arrive. We ask whether a single frame, read carefully, carries the same structure, and find that it does, provided no individual cue is trusted alone. We reframe pseudo-label construction as a problem of cross-modal agreement: frozen semantic codes propose categories, monocular depth proposes physical separations, and a guarded filter accepts a label only when both cues concur. Two compact mechanisms instantiate this principle inside an otherwise standard pipeline. DCFA is a 40K-parameter zero-initialized residual that bends a frozen 90-dimensional CAUSE-TR code toward plausible geometry through a 16-dimensional sinusoidal depth embedding; SIMCF-ABC enforces semantic uniformity inside each instance proposal, merges adjacent depth fragments only when class identity and DINOv3 cosine similarity agree ($\tau = 0.85$), and rejects pixels whose depth violates per-class statistics. On Cityscapes, DCFA lifts semantic clustering mIoU from 52.69 to 55.29, and the full pipeline raises pseudo-label panoptic quality from 24.54 to 25.85, with the gain concentrated on the things split ($PQ_{th} \ 12.31 \rightarrow 14.70$). Trained with the published CUPS bootstrapping and self-training recipe on a frozen DINOv3 ViT-B/16 Cascade Mask R-CNN, the resulting detector reaches 35.83 PQ on the Cityscapes validation set, against 27.80 PQ for the published CUPS configuration (ResNet-50, stereo+motion) and 32.76 PQ for the same recipe trained on monocular pseudo-labels without our cross-modal filter; the +3.07 PQ contribution-isolated gap names what monocular agreement adds under identical training.

Monocular Cross-Modal Agreement for Unsupervised Panoptic Segmentation

Anonymous CVPR submission

1 Introduction

Panoptic segmentation assigns a category to every pixel and a distinct identity to every countable object. The task underpins autonomous driving, robotic manipulation, and dense scene parsing, and its supervised counterpart sits at the top of every benchmark that supplies dense human masks. Producing those masks is the cost. Cityscapes-style annotation requires a pixel-level mask for every visible instance, and the cost compounds with each new dataset and domain. The unsupervised setting asks whether a model can recover panoptic structure from raw pixels alone, and recent progress has begun to answer.

The progress has come from three lineages. Self-supervised semantic discovery on top of frozen DINO/DINOv2 features [STEGO, CAUSE] now yields region-level partitions coherent enough to serve as semantic pseudo-labels. Cut-and-learn instance discovery [LOST, TokenCut, MaskCut, CutLER] extracts foreground masks from the same SSL features and bootstraps detectors from them; CutS3D pivots toward scene-centric data by lifting depth to a 3-D point cloud and applying LocalCut to disambiguate adjacent instances. CUPS [Hahn et al., 2025] is the strongest scene-centric assembly of these threads: it constructs panoptic pseudo-labels from a CAUSE-style semantic head, stereo-derived depth, and unsupervised scene flow, then trains a Cascade Mask R-CNN with DropLoss bootstrapping and EMA self-training, setting the current published Cityscapes bar.

The recipe pays for that score with a hardware commitment. CUPS requires calibrated stereo at pseudo-label time and unsupervised optical flow over short video clips; both come from sensor configurations that scene-centric datasets often lack. Most images in the wild arrive as a single monocular frame, captured from a phone, a dashcam, or a single calibrated camera, with neither a paired stereo view nor a dense temporal neighbourhood. A method that depends on this multi-view input narrows unsupervised panoptic segmentation to a small slice of available imagery. The question worth asking is whether the multi-view input is fundamental, or whether a single frame, read carefully, carries the same structure.

We answer that the multi-view input is not fundamental. Frozen monocular priors — a self-supervised seman-

tic code, a monocular depth estimator, and dense self-supervised features for cross-region matching — together expose enough structure to take stereo and motion’s place at pseudo-label time, provided no individual cue is asked to do too much. Frozen semantic codes are appearance-driven and routinely merge two adjacent same-class objects into one region while splitting a single physical surface across an illumination change. Monocular depth recovers physical separations between adjacent objects but over-fragments single surfaces along intra-object material seams. Dense feature similarity can heal some depth fragmentations, but only when constrained by category. The natural framing is cross-modal agreement: depth gradients propose object boundaries; they do not identify them. Geometry proposes the split, semantics and appearance decide whether the split is credible. A pseudo-label is reliable when independent monocular cues with different failure modes converge on the same answer.

Instantiating this principle inside an otherwise standard pipeline lifts pseudo-label panoptic quality from 24.54 to 25.85 on Cityscapes, with the gain concentrated on the things split ($PQ_{th} 12.31 \rightarrow 14.70$). Trained with the published CUPS bootstrapping and three rounds of EMA self-training on a frozen DINOv3 ViT-B/16 Cascade Mask R-CNN, the resulting detector reaches 35.83 PQ on Cityscapes val. The published CUPS configuration (ResNet-50 backbone, stereo + motion at pseudo-label time) reaches 27.80 PQ on the same benchmark; the same recipe trained on monocular pseudo-labels without our cross-modal filter reaches 32.76 PQ. The +3.07 PQ contribution-isolated gap names what monocular agreement adds to monocular pseudo-labels under identical Stage-2 training.

We make three contributions:

- Stereo-and-motion-free pseudo-label generation for scene-centric panoptic segmentation. Monocular semantic appearance and monocular depth, composed by an agreement filter, supply enough structure for scene-centric unsupervised panoptic-label generation, removing CUPS’s stereo + motion requirement at pseudo-label time.
- Two compact mechanisms for cross-modal agreement. A depth-conditioned residual adapter (DCFA) bends a frozen 90-dimensional semantic code toward depth-consistent regions and lifts semantic

mIoU from 52.69 to 55.29; a three-stage label filter (SIMCF-ABC) accepts, revises, or rejects candidate labels through semantic, geometric, and appearance agreement and contributes the bulk of the +1.31 PQ pseudo-label gain.

- 35.83 PQ on Cityscapes under the inherited CUPS recipe. Trained on these monocular pseudo-labels with a frozen DINOv3 ViT-B/16 Cascade Mask R-CNN, the resulting detector reaches 35.83 PQ on Cityscapes val, with a +3.07 PQ contribution-isolated gap against the same-backbone monocular baseline.

Section 2 surveys the unsupervised semantic, instance, and panoptic literatures that frame our setting. Section 3 develops the four pipeline modules. Section 4 reports the headline result, the contribution-isolated ablations, and cross-dataset transfer. Section 5 discusses the principle established by these results, and Sections 6 and 7 catalog limitations and broader impact.

2 Related Work

2.1 Unsupervised Semantic Segmentation

Early unsupervised semantic segmentation methods optimized clustering objectives over learned image representations: IIC [Ji et al., 2019] maximized mutual information between augmented views, and PiCIE [Cho et al., 2021] enforced photometric and geometric invariance to recover pixel-level groupings. The arrival of self-supervised vision transformers [Caron et al., 2021; Oquab et al., 2024] reshaped the field, with dense ViT features clustering cleanly enough that distillation methods became dominant. STEGO [Hamilton et al., 2022] distilled DINO correspondences into compact dense embeddings; HP [Seong et al., 2023] improved this contrastive pool by discovering hidden positives; CAUSE [Kim et al., 2024] structured the latent space through a concept clusterbook, producing a 90-dimensional semantic code. Recent variants refine the clustering objective on top of these SSL features but do not change what is being clustered.

Monocular depth has been considered as an auxiliary signal throughout this lineage. DepthG [Sick et al., 2024] adds a depth-feature correlation loss together with depth-guided positive/negative sampling on top of a STEGO-style backbone, while CUPS [Hahn et al., 2025] uses depth as a contrastive auxiliary in its DINO-distillation semantic head. Earlier work used depth as a multi-task pre-training signal [Hoyer et al., 2021]. Unlike these prior unsupervised depth-guided methods that use depth as a loss-side signal or sampling prior, DCFA conditions a frozen unsupervised semantic code on monocular depth through a small residual adapter, leaving the backbone untouched.

2.2 Unsupervised Instance Segmentation

The dominant line of unsupervised instance discovery applies graph-cut-style partitioning to self-supervised features. LOST [Siméoni et al., 2021] localized a single salient object per image; TokenCut [Wang et al., 2023a] generalized this through normalized cuts on DINO patch tokens; MaskCut [Wang et al., 2023b] iterated TokenCut to extract multiple masks per image; CutLER [Wang et al., 2023c] turned MaskCut pseudo-masks into supervision for a class-agnostic Cascade Mask R-CNN, establishing the cut-and-learn template. MaskDistill [Van Gansbeke et al., 2022] and FreeSOLO [Wang et al., 2022] explored alternative pseudo-mask formulations within the same paradigm.

Recent extensions diversify the cut-and-learn template. CuVLER [Arica et al., 2024] strengthens MaskCut with multi-SSL feature voting and soft-target distillation; COLER [Feng et al., 2025] cuts in a single pass without k-means; ProMerge [Li et al., 2024] introduces prompt-and-merge grouping with background-based pruning; unMORE [Yang et al., 2025] learns existence/center/boundary fields; UnSAM and UnSAMv2 [Wang et al., 2024; Yu et al., 2025] add granularity-controlled segmentation on SA-1B-scale data. Slot-attention models such as DINOSAUR [Seitzer et al., 2023] and its motion-refined extension MR-DINOSAUR [Gong et al., 2025] are adjacent alternatives that largely target object-centric or class-agnostic instance discovery rather than the instance branch of scene-centric panoptic pseudo-labeling.

A separate subtitle injects geometric information explicitly, and CutS3D [Sick et al., 2025] is the closest foil to our instance pipeline. CutS3D lifts each image to a 3-D point cloud through monocular depth, constructs a 3-D k-NN graph, and applies LocalCut to disambiguate adjacent instances that 2-D NCut merges. We adopt CutS3D’s monocular premise but stay entirely in 2-D: depth gradients provide a 2-D proxy for the separation cue CutS3D obtains after unprojecting depth to 3-D, namely local discontinuities between physically distinct surfaces. After Sobel filtering and connected-component analysis within thing-eligible semantic regions, this 2-D treatment supplies the instance side of our pseudo-labels without lifting to a point cloud or constructing a 3-D graph.

2.3 Unsupervised Panoptic Segmentation

Unsupervised panoptic segmentation is a young task. The most directly comparable Cityscapes results to date come from three works. U2Seg [Niu et al., 2024] is the first unified framework, composing STEGO and MaskCut on object-centric data and transferring to scene-centric Cityscapes; it inherits the dominant-foreground bias of MaskCut and reaches roughly 18 PQ. CUPS [Hahn et al., 2025] is the current scene-centric state of the art and con-

structs pseudo-labels from pseudo-label-time stereo and motion cues — stereo video, unsupervised optical flow [Stone et al., 2021], and SF2SE3 scene-flow segmentation [Sommer et al., 2022] — then trains a Cascade Mask R-CNN with DropLoss bootstrapping and three rounds of EMA self-training. S2-UniSeg [Xu et al., 2025] is a concurrent monocular alternative that performs well but relies on large-scale segmentation-oriented pretraining on SA-1B before adaptation. Our setting differs from these concurrent monocular alternatives by generating pseudo-labels from target-dataset monocular frames alone, without stereo/video capture, explicit 3-D reconstruction, or SA-1B-scale segmentation pretraining. We inherit CUPS-style detector training, DropLoss bootstrapping, copy-paste augmentation, and EMA self-training, and replace only the pseudo-label generator.

3 Method

3.1 Problem Setup and Pipeline Overview

We address scene-centric unsupervised panoptic segmentation in the regime introduced by CUPS [Hahn et al., 2025], where a model must produce dense panoptic predictions without human annotations. Concretely, training is given an unlabeled image set $\mathcal{D}_{\text{train}} = \{x_i\}_{i=1}^N$, in which each frame $x_i \in \mathbb{R}^{H \times W \times 3}$ is a single RGB image drawn from the target dataset, here Cityscapes [Cordts et al., 2016]. The setting provides no semantic, instance, or panoptic annotations, no ground-truth or sensor-provided depth maps, no calibrated stereo views, and no temporal neighbours. We use only unlabeled target images for pseudo-label construction and detector training, while relying on frozen off-the-shelf visual and depth priors. Formally, the two stages can be written as

$$G(x) = (\hat{S}_x, \hat{I}_x), \quad F(x) = (S_x, I_x), \quad (1)$$

where G is the monocular pseudo-label generator used during training and F is the panoptic detector evaluated at test time. The maps $\hat{S}_x$ and S_x assign semantic labels to pixels, while $\hat{I}_x$ and I_x assign instance identities to pixels belonging to countable object classes.

Working strictly within this monocular regime, we adopt the same two-stage decomposition that organizes CUPS. Stage 1 fits any pseudo-labeling components using only unlabeled target images, then applies the resulting generator independently to each frame, turning every x_i into training targets $(\hat{S}_{x_i}, \hat{I}_{x_i})$. Stage 2 trains a panoptic detector on these pseudo-labels with a fixed recipe inherited from CUPS. The decomposition is deliberate: by holding Stage 2 constant, changes in final detector performance can be attributed to the pseudo-label source under the fixed Stage-2 recipe, which is the locus of our contributions and the only place where stereo and motion cues entered the original CUPS recipe.

What replaces those cues is a Stage-1 generator built from three modules, illustrated in Figure 1. A semantic branch produces $\hat{S}_x$ by passing x through a frozen DINOv2 backbone followed by the frozen CAUSE-TR head [Kim et al., 2024], whose semantic code is refined by the Depth-Conditioned Feature Adapter (DCFA) before being clustered into pseudo-classes. An instance branch produces $\hat{I}_x$ by computing Sobel responses on a monocular depth map and extracting connected components inside thing-eligible semantic regions. A cross-modal filter, SIMCF-ABC, then accepts, revises, or rejects candidate labels using three agreement cues: semantic identity, instance geometry, and feature similarity. The accepted pseudo-labels are passed to the Stage-2 trainer.

These three modules are developed in turn in the remainder of this section. Section 3.2 defines the monocular baseline that replaces CUPS’s stereo and motion pseudo-labels with depth-gradient connected components while keeping detector training fixed. Section 3.3 introduces DCFA, which conditions the frozen semantic code on monocular depth through a residual feature-level injection. Section 3.4 specifies the SIMCF-ABC filter and the agreement criteria each cue contributes. Section 3.5 details the Stage-2 detector, the inherited DropLoss bootstrapping, copy-paste augmentation, and EMA self-training schedule, and the inference path that converts a trained detector into a panoptic prediction at test time.

3.2 Monocular Pseudo-Label Baseline

Stereo and scene flow give CUPS access to physical-separation cues at pseudo-label time: a 3-D boundary between two adjacent surfaces produces a discontinuity in disparity or in inter-frame motion that separates one instance from another. Monocular depth exposes many of the same physical-separation cues in a single frame, provided that the discontinuity is read at the right spatial scale and only within a region whose semantic identity has already been settled. The Stage-1 generator G from Eq. 1 is built around this observation: a frozen semantic branch produces $\hat{S}_x$, an instance branch produces $\hat{I}_x$ from depth-gradient cuts taken inside thing-eligible semantic regions, and the resulting pair is passed to the unchanged Stage-2 trainer.

The semantic branch is assembled entirely from frozen priors so that the boundary signal it supplies to the instance branch is reproducible across frames. Each image x is encoded by a frozen DINOv2 ViT-B/14 backbone [Oquab et al., 2024], whose dense patch features are then reduced by a frozen CAUSE-TR head [Kim et al., 2024] to a 90-dimensional semantic code. We L2-normalize these codes (so that centroid distance is scale-invariant to feature magnitude) and partition them with k-means on the unlabeled training set into $k = 80$ centroids, assigning each pixel the index of its nearest centroid to obtain $\hat{S}_x$.

The cardinality $k = 80$ is deliberately well above the 19-class evaluation taxonomy: collapsing to 19 classes at this stage would erase fine visual modes that the depth-based instance branch later relies on, such as occluded versus unoccluded vehicles. CUPS reports that this regime monotonically improves panoptic quality as k exceeds the evaluation cardinality [Hahn et al., 2025]. We cluster the CAUSE-TR code rather than the raw 768-dimensional DINOv2 feature because CAUSE-TR has already organized the manifold into a concept-coherent partition; clustering on the raw feature would re-mix appearance modes that the unsupervised code has already factored out.

With $\hat{S}_x$ in hand, the instance branch reads physical-separation structure from a single monocular depth map $D \in \mathbb{R}^{H \times W}$. For the detector results reported in the main experiments we use a frozen DepthPro network [Bochkovskii et al., 2024], chosen because it balances boundary sharpness, which the gradient operator below requires, against surface smoothness, which keeps the interior of a single object connected. We apply 3×3 Sobel kernels K_x, K_y to D and form the per-pixel gradient magnitude $\|\nabla D(p)\| = \sqrt{(K_x * D)^2(p) + (K_y * D)^2(p)}$, since a physical 3-D boundary projects to a depth discontinuity that classical gradient operators are designed to localize. Pixels with $\|\nabla D(p)\| > \tau_d = 0.01$ are flagged as candidate boundaries; the threshold rejects high-frequency fluctuations that monocular networks produce as quantization noise. Connected-component labelling with 8-neighbour adjacency (suitable for the diagonal boundaries that depth gradients produce at oblique surfaces) is then run only inside thing-eligible semantic regions, defined as over-clusters whose dominant Cityscapes class is one of the eight countable categories (car, truck, bus, train, motorcycle, bicycle, person, rider) [Cordts et al., 2016]. This restriction is important because depth gradients fire not only at object boundaries but also at intra-object material seams such as the hood-windshield transition on a car, and confining components to the interior of a single semantic region lets the semantic cue absorb such intra-class splits while still permitting depth to separate adjacent same-class objects. Components below $A_{\min} = 1000$ pixels are discarded as residual depth artifacts unlikely to form reliable detector targets at our working resolution, and the surviving components are dilated with three iterations of a 3×3 structuring element, which partially restores boundary pixels removed by thresholding.

Composing the two branches yields the baseline pseudo-label pair $(\hat{S}_x, \hat{I}_x)$, which is the monocular-depth baseline instantiation of G from Eq. 1 and the input to the inherited Stage-2 trainer. Because the fitted generator is applied deterministically to each frame, the failure modes of this baseline are also localized: the depth-driven instance branch behaves correctly wherever a 3-D surface change is visible, but the semantic regions inherited from the frozen CAUSE-TR code are appearance-driven and

can split a single physical surface across an illumination change, such as sun and shade on the same road, or merge two adjacent same-class objects whose appearance differs only in fine geometry. Closing this gap requires conditioning the semantic code on the same depth signal that already governs the instance branch, which motivates the Depth-Conditioned Feature Adapter introduced in Section 3.3.

3.3 Depth-Conditioned Feature Adapter

The semantic regions inherited from the frozen CAUSE-TR code are appearance-driven, and Section 3.2 closed by naming the two failure modes this produces inside G : a single physical surface is split across an illumination change, such as sun and shade on the same road, and two adjacent same-class objects whose appearance differs only in fine geometry are merged into one. Both expose cases where the depth signal that already governs the instance branch can disambiguate appearance-driven semantic errors, but only if the semantic code can be conditioned on D without retraining the backbone that the rest of the pipeline depends on. We therefore introduce a small residual adapter that consumes the depth value d_u at pixel u and adds a learned depth-driven shift to the frozen 90-dimensional code z_u , nudging cluster boundaries toward depth-consistent regions where appearance fails, while preserving the unsupervised prior elsewhere.

Conditioning on d_u requires a representation rich enough for a small MLP to read both fine boundary differences and coarse depth ordering. Since $d_u \in [0, 1]$ is a scalar with low dynamic range, we lift it through a 16-dimensional sinusoidal positional encoding with eight exponentially-spaced frequencies $\omega_k = 2^k \pi$ for $k = 0, \dots, 7$,

$$e(d_u) = [\sin(\omega_0 d_u), \cos(\omega_0 d_u), \dots, \sin(\omega_7 d_u), \cos(\omega_7 d_u)] \in \mathbb{R}^{16}, \quad (2)$$

so that the high-frequency channels make boundary-scale depth differences linearly readable while the low-frequency channels expose foreground-versus-background ordering.

The encoding $e(d_u)$ is then passed through a two-layer MLP $r : \mathbb{R}^{16} \rightarrow \mathbb{R}^{90}$ with hidden width $h = 384$ (a width chosen to give the residual enough capacity at the 90-dimensional output), ReLU activation, and a zero-initialized output projection. The MLP produces a depth-only shift that is added residually to the frozen code,

$$\tilde{z}_u = z_u + r(e(d_u)), \quad (3)$$

so that depth modulates the frozen semantic code through an additive residual. The construction has approximately 40K trainable parameters, small relative to the roughly 86M parameters of the frozen ViT backbone, so the adapter acts as a calibration on the prior rather than a re-training of it. Two properties of this construction matter

for the rest of the pipeline. Because the output projection is zero-initialized, $\tilde{z}_u = z_u$ at the first optimizer step, so the adapter cannot degrade the frozen prior before any depth signal has been read. And because the DINOv2 backbone and CAUSE-TR head are kept frozen, the same $\tilde{z}_u$ remains usable by the k-means partition of Section 3.2 and by the feature-similarity cue used in SIMCF-ABC (Section 3.4).

Training the adapter calls for a self-supervised signal that pulls the adapted codes of pixels at similar depth toward each other while keeping them on the original CAUSE-TR manifold. DepthG [Sick et al., 2024] established that depth-feature correlation can guide unsupervised semantic learning when depth governs the loss, and we reuse this principle, but apply it at the adapter level rather than to the backbone weights. For each minibatch image, we sample a set $\mathcal{U}$ of $|\mathcal{U}| = 1024$ random pixels and form a pair set $\mathcal{P}$ with $|\mathcal{P}| = P = 1024$ pixel pairs (i, j) drawn from $\mathcal{U}$ (a stochastic-estimator scale that gives a low-variance gradient on the 32×64 feature grid), with depth-similarity weights $w_{ij} = \exp(-(d_i - d_j)^2 / (2\sigma_d^2))$ and $\sigma_d = 0.5$. The training objective combines a depth-guided correlation loss $\mathcal{L}_{\text{depth}}$ in the sense of DepthG with a preservation loss $\mathcal{L}_{\text{preserve}}$ that anchors the adapter to the frozen prior,

$$\mathcal{L}_{\text{DCFA}} = \underbrace{\frac{1}{P} \sum_{(i,j) \in \mathcal{P}} w_{ij} (1 - \cos(\tilde{z}_i, \tilde{z}_j))^2}_{\mathcal{L}_{\text{depth}}} + \underbrace{\lambda_{\text{preserve}} \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \|\tilde{z}_u - z_u\|_2^2}_{\mathcal{L}_{\text{preserve}}} \quad (4)$$

with $\lambda_{\text{preserve}} = 20$. $\mathcal{L}_{\text{depth}}$ encourages pixels on the same physical surface to remain clustered even when appearance changes, while $\mathcal{L}_{\text{preserve}}$ anchors the adapter to the original CAUSE-TR manifold and prevents collapse onto a degenerate point in regions where depth alone is uninformative. We set $\lambda_{\text{preserve}} = 20$ as a value high enough that the preservation term dominates outside high-similarity neighbourhoods. This contrasts with DepthG, which injects the depth-correlation signal into the backbone’s training loop and modifies the ViT weights end-to-end. Freezing the backbone keeps the prior reusable across Section 3.2’s clustering and Section 3.4’s similarity cue, which is the property we exploit downstream.

Even with depth-conditioned semantic codes, the pseudo-label pair $(\hat{S}_x, \hat{I}_x)$ can still disagree at the output level. A depth fragment may straddle the hood-windshield seam of a car despite a coherent semantic region, and a pixel may take a depth value far from the per-class statistics of the class it has been assigned. SIMCF-ABC (Section 3.4) addresses these output-level disagreements through a label filter that requires three complementary cues — semantic identity, instance geometry, and feature similarity — to agree before a candidate label is accepted.

3.4 SIMCF-ABC: Cross-Modal Agreement Filter

The output-level disagreements named at the close of Section 3.3 cannot be reached by a feature-level conditioner: once the semantic and instance branches of G have produced a candidate pair (S_x^0, I_x^0) , a fragment may still straddle an intra-object material seam, a per-pixel class assignment may still sit at a depth far from the empirical statistics of its class, and a single object may still be carved into pieces by a depth gradient that runs through it. We therefore introduce SIMCF-ABC, the Semantic-Instance Mutual Cross-modal Filter, the final stage inside G that turns the raw candidate (S_x^0, I_x^0) into the filtered pseudo-labels $(\hat{S}_x, \hat{I}_x)$ that the Stage-2 trainer consumes. SIMCF-ABC checks each candidate against three complementary cues, accepting, revising, or rejecting every label according to whether semantic identity, instance geometry, and feature similarity agree. Stage A revises pixels whose semantics disagree with the dominant class of their instance, stage B merges adjacent fragments whose appearances coincide, and stage C rejects pixels whose depth value is incompatible with the per-class depth distribution. Algorithm 1 sketches the three stages applied in sequence.

The first cue, semantic uniformity, asks whether the interior of an instance proposal can settle on one pseudo-class. Let ϕ be a pseudo-class grouping that maps each of the 80 over-clusters to a smaller pseudo-class index. The grouping is fit once on the 80 over-cluster centroids from Section 3.2 by a coarser k-means, without consulting any target annotations, so ϕ remains part of the same unsupervised pipeline as the over-clusters themselves. For each instance proposal k with pixel set I_k , the dominant pseudo-class is

$$c_k^* = \arg \max_c |\{p \in I_k : \phi(S_x^0(p)) = c\}|, \quad (5)$$

and every pixel $p \in I_k$ with $\phi(S_x^0(p)) \neq c_k^*$ is reassigned to the most common c_k^* -consistent over-cluster

$$s_k^* = \arg \max_s |\{p \in I_k : S_x^0(p) = s \wedge \phi(s) = c_k^*\}|. \quad (6)$$

An instance whose interior cannot settle on a single pseudo-class provides unreliable supervision for the Stage-2 detector; SIMCF-A therefore aligns the semantic and instance maps inside each region without consulting the cues that follow, treating the filter as a structural guard rather than a discard rule.

The second cue, feature similarity, recognizes that depth fragmentation often breaks single objects along intra-object material seams of the kind named in Section 3.2 — the hood-windshield seam on a car being the canonical case. For each surviving proposal k we compute the L2-normalized mean DINOv3 ViT-B/16 patch feature

$$\hat{f}_k = \bar{f}_k / \|\bar{f}_k\|_2, \quad \bar{f}_k = \frac{1}{|\bar{I}_k|} \sum_{p \in \bar{I}_k} v(p), \quad (7)$$

where $v(p) \in \mathbb{R}^{768}$ is the DINOv3 patch feature at the 32×64 grid resolution and $\tilde{I}_k$ is the patch-resolution restriction of I_k . Two proposals a, b are adjacent iff $\text{dilate}(I_a, r=3) \cap I_b \neq \emptyset$, where the small radius keeps the adjacency relation local rather than global. They are merged iff their dominant pseudo-classes agree and their cosine similarity exceeds $\tau_{\text{sim}} = 0.85$, a value that requires near-identical DINOv3 appearance evidence so that the merge happens only between fragments whose interior content is barely distinguishable. Transitive merges are resolved by union-find. Feature similarity supplies a complementary appearance prior that the depth and semantic cues cannot, recovering single-object identity in exactly the case where a depth gradient has produced a spurious cut.

The third cue, per-pseudo-class depth statistics, treats the depth map D as a sanity check on the assignments that survive A and B. For each pseudo-class c we form the empirical mean μ_c and standard deviation σ_c over all pixels currently labelled c , with σ_c floored by a small ϵ and pseudo-classes containing too few pixels skipped to avoid aggressive rejection in degenerate cases. Defining the dimensionless multiplier $\eta = 2.5$, we reject any pixel p whose depth deviates from its assigned-pseudo-class mean by more than $\eta\sigma_c$, that is $|D(p) - \mu_{c(p)}| > \eta\sigma_{c(p)}$, by assigning the void label and removing the pixel from both semantic and instance supervision. The multiplier is wide enough to absorb legitimate within-pseudo-class depth spread; a ground-plane-like pixel observed at far-background depth, or a sky-like pixel observed at near-foreground depth, is much more plausibly a labelling error than a rare physical configuration, and rejecting these outliers prevents the Stage-2 detector from absorbing depth-implausible per-pseudo-class priors before training begins.

SIMCF-ABC trains no parameters of its own; within the Stage-1 pseudo-label generator G , DCFA from Section 3.3 is the only learned component. The filtered labels $(\hat{S}_x, \hat{I}_x)$ are then passed to the Stage-2 trainer, which is the subject of Section 3.5.

3.5 Stage-2 Detector Training and Panoptic Inference

Filtering completes the Stage-1 generator G . The resulting pseudo-labels $(\hat{S}_x, \hat{I}_x)$ provide the supervision for Stage 2, while the RGB image x remains the detector input. We keep the CUPS-style detector training recipe fixed across all configurations [Hahn et al., 2025], including its loss, copy-paste augmentation, bootstrapping schedule, and EMA self-training. This lets the experiments isolate the effect of the pseudo-label source: differences in detector performance can be attributed primarily to the pseudo-label source under the fixed Stage-2 recipe. Stage 2 is also where the pipeline acquires generalization beyond per-image pseudo-labels, since a single detector trained across the full training set can absorb some per-

Algorithm 1 SIMCF-ABC filter, applied per image

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Require: Candidate pair  $(S_x^0, I_x^0)$ , depth  $D$ , grouping  $\phi$ , DINOv3
features  $v$ , thresholds  $\tau_{\text{sim}}, \eta, \epsilon$ 
Ensure: Filtered pseudo-labels  $(\hat{S}_x, \hat{I}_x)$ 
# SIMCF-A — semantic uniformity
1: for each instance  $k$  with pixel set  $I_k$  do
2:    $c_k^* \leftarrow \arg \max_c |\{p \in I_k : \phi(S_x^0(p)) = c\}|$ 
3:    $s_k^* \leftarrow \arg \max_s |\{p \in I_k : S_x^0(p) = s \wedge \phi(s) = c_k^*\}|$ 
4:   reassign every  $p \in I_k$  with  $\phi(S_x^0(p)) \neq c_k^*$  to  $s_k^*$ 
5: end for
# SIMCF-B — feature-similarity merge of adjacent fragments
6: for each instance  $k$  do
7:    $\bar{f}_k \leftarrow \bar{f}_k / \|\bar{f}_k\|_2$  where  $\bar{f}_k = \frac{1}{|I_k|} \sum_{p \in I_k} v(p)$ 
8: end for
9: for each pair of adjacent instances  $(a, b)$  do
10:  if  $c_a^* = c_b^*$  and  $\langle \bar{f}_a, \bar{f}_b \rangle > \tau_{\text{sim}}$  then
11:    Merge  $I_a$  and  $I_b$  by union-find
12:  end if
13: end for
# SIMCF-C — per-class depth-statistics rejection
14: for each pseudo-class  $c$  with sufficient pixel count do
15:  Estimate  $(\mu_c, \sigma_c)$  from  $D$  over pixels labelled  $c$ 
16:   $\sigma_c \leftarrow \max(\sigma_c, \epsilon)$ 
17: end for
18: for every surviving pixel  $p$  do
19:  if  $|D(p) - \mu_{c(p)}| > \eta\sigma_{c(p)}$  then
20:    Assign void to  $p$  (in both  $\hat{S}_x$  and  $\hat{I}_x$ )
21:  end if
22: end for
23: return  $(\hat{S}_x, \hat{I}_x)$ 

```

image pseudo-label noise through shared parameters and self-training.

The trainer instantiates a Cascade Mask R-CNN [Cai and Vasconcelos, 2018] with three cascade stages on top of a frozen DINOv3 ViT-B/16 backbone [Siméoni et al., 2025]. We freeze the backbone to keep the detector capacity fixed across pseudo-label configurations and to prevent backbone adaptation from confounding pseudo-label quality, mirroring the frozen-prior treatment used in the Section 3.2 pseudo-label generator and the Section 3.3 adapter. Inside the detector, the cascade heads, region proposal network, and panoptic prediction layers are trained on $(\hat{S}_x, \hat{I}_x)$, taking x as input, with the CUPS-style loss formulation.

Two ingredients of the inherited recipe interact directly with the void labels produced by SIMCF-ABC. Following CUPS, DropLoss [Wang et al., 2023] masks losses on void or unreliable regions so that rejected pseudo-labels are not treated as hard negatives, making void a soft abstention rather than a class signal. Self-enhanced copy-paste augmentation operates at the input level: at each training iteration, instances from the pseudo-labels of one image are pasted into another image at random positions, and their semantic and instance labels are carried over into the destination supervision. This augmentation increases exposure to thing-class instances, which would otherwise be sparse on Cityscapes, and is taken from the same CUPS specification as the loss.

Once the cascade trainer reaches its bootstrapped checkpoint, three rounds of EMA self-training continue from it. In each round, an exponential-moving-average teacher is maintained over the student weights and used to relabel the unlabeled training set; teacher predictions are thresholded by the confidence rule specified by CUPS, and the student is re-trained on the resulting refined pseudo-labels. The teacher’s role is to provide a smoother, lower-variance label distribution than G alone supplies, in particular for thing-class boundaries where Stage-1 labels are limited by monocular depth discontinuities. The detector F in Eq. 1 is the student at the end of Round 3.

At test time, F takes a single RGB image x and emits a dense semantic prediction over the pseudo-class vocabulary used for Stage-2 supervision together with a set of instance proposals from the cascade heads. A panoptic merge step combines them into (S_x, I_x) along the lines of [Kirillov et al., 2019]: pixels assigned to thing-eligible pseudo-classes inherit the instance identifier of the highest-confidence instance proposal that covers them, pixels assigned to stuff pseudo-classes receive instance identifier 0, and overlapping instance proposals on the same pixel are resolved by confidence ranking. The pseudo-class outputs are mapped to benchmark classes only at evaluation time. Ground-truth annotations are used only for evaluation, including the Hungarian mapping from pseudo-classes to benchmark classes; they are never used to train G or F .

4 Experiments

We evaluate the proposed pipeline against three claims: monocular pseudo-labels seeded into the CUPS bootstrapping recipe produce a trained Cityscapes detector that exceeds the published CUPS baseline; DCFA and SIMCF-ABC correct different error modes inside the pseudo-label generator; and the trained detector generalises within the driving manifold but is bounded by its fixed Cityscapes-derived class space. Two evaluation regimes structure every table: pseudo-label evaluations measure G ’s output before any panoptic model is trained, and trained-model evaluations measure the Cascade Mask R-CNN that emerges from CUPS-style bootstrapping and self-training. Treating them separately prevents pseudo-label quality from being conflated with detector capacity, and prevents a strong backbone from rescuing weak supervision.

4.1 Experimental Setup

Pseudo-label generation and detector training use only Cityscapes train [Cordts et al., 2016] (2,975 images, no annotations); evaluation reports Cityscapes val (500 images) under the 27-class CAUSE pseudo-vocabulary with a Hungarian map to Cityscapes’ 19 evaluation classes

[Kirillov et al., 2019], following CUPS. The Stage-1 generator G uses a frozen DINOv2 ViT-B/14 backbone [Oquab et al., 2024], the frozen CAUSE-TR head [Kim et al., 2024], DCFA ($h = 384$, 16-D sinusoidal depth embedding, $\lambda_{\text{preserve}} = 20$, $\sim 40\text{K}$ trainable parameters trained for 10K iterations with AdamW at learning rate 10^{-3}), k -means over-clustering at $k = 80$, the Section 3.2 monocular instance branch with $A_{\text{min}} = 1000$ and three iterations of dilation, and the SIMCF-ABC filter ($\tau_{\text{sim}} = 0.85$, $r = 3$, $\eta = 2.5$). DepthPro [Bochkovskii et al., 2024] supplies the monocular depth map. The detector F is a Cascade Mask R-CNN [Cai and Vasconcelos, 2018] with three cascade stages on a frozen DINOv3 ViT-B/16 backbone [Siméoni et al., 2025], trained under the CUPS recipe: DropLoss [Wang et al., 2023] bootstrapping for 8K steps, self-enhanced copy-paste augmentation, and three rounds of EMA self-training (5K steps each, EMA coefficient 0.999, multi-view teacher with three scales). All Cityscapes numbers report a single seed; total wall-clock time on a single NVIDIA RTX A6000 (48 GB) is approximately 24 hours. The detector training uses $\tau_d = 0.20$ for the depth-gradient threshold rather than the standalone-optimum $\tau_d = 0.01$ from the depth-instance ablation; Section 4.3 explains this constraint.

4.2 Main Cityscapes Results

Table 1 records the central comparison on Cityscapes val. The Regime column groups rows by what is being measured: a published trained-model baseline, our raw monocular supervision before any training, and our trained model after Stage-2 bootstrapping plus three rounds of self-training.

The claim we make is therefore narrower than the headline. Monocular pseudo-labels reach 25.85 PQ — close enough to seed a strong CUPS-style recipe — and once seeded, that recipe produces a trained model that exceeds the published CUPS number by 8.03 PQ. Whether the gap is principally pseudo-label quality, backbone capacity, or detector implementation cannot be decided from Table 1 alone, and Section 6 names a same-backbone CUPS rerun as the most decisive missing control.

The Stage-3 trajectory on Cityscapes val is monotonic across self-training steps: 31.87 PQ at step 800, 33.78 at step 1000, 35.47 at step 2200, 35.83 at step 3000. The shape is consistent with self-training densifying the structure that G supplies rather than memorizing the per-image label distribution. The most striking row-to-row movement is the things-class lift: pseudo-label $\text{PQ}_{\text{th}} = 14.70$ rises to trained-model $\text{PQ}_{\text{th}} = 36.26$, a +21.56 PQ_{th} gain that the inherited CUPS recipe extracts from supervision G alone could not match. Self-training is a CUPS-recipe ingredient, not a contribution of this paper, but it is load-bearing for the headline number.

The mIoU column inverts the usual training-improves-mIoU pattern (56.22 at the pseudo-label row, 44.56 at the

Table 1: Main results on Cityscapes val. Rows are not a head-to-head scoreboard: the CUPS row reports a published trained model under a different backbone and training implementation, not the CUPS pseudo-label cost. Our pseudo-label row alone reaches 25.85 PQ, below CUPS’s 27.80; the +8.03 PQ headline gap (35.83 vs 27.80) materializes only after CUPS-style bootstrapping with DINOv3 Cascade Mask R-CNN.

Regime	Method	Backbone / training	PQ	PQ _{th}	PQ _{st}	mIoU
Trained baseline	CUPS published [Hahn et al., 2025]	published CUPS	27.80	17.70	35.10	26.80
Pseudo-labels (ours)	DCFA + DepthPro + SIMCF-ABC	none	25.85	14.70	33.96	56.22
Trained model (ours)	DCFA + DepthPro + SIMCF-ABC	DINOv3 Cascade, Stage 3	35.83	36.26	35.56	44.56

Table 2: Component complementarity at fixed $\tau_d = 0.20$.

Variant	DCFA	SIMCF	PQ	PQ _{th}	PQ _{st}	mIoU
Raw $K=80$ + DepthPro	–	–	24.54	12.31	33.43	56.56
DCFA only	✓	–	25.22	13.16	33.99	56.16
SIMCF-ABC only	–	✓	25.27	13.64	33.73	56.57
DCFA + SIMCF-ABC	✓	✓	25.85	14.70	33.96	56.22

Table 3: Depth-source comparison at standalone-optimum τ per estimator.

Depth source	Type	τ	PQ	PQ _{th}	PQ _{st}
No depth (CC-only)	none	n/a	24.80	14.93	32.08
SPIddepth	self-sup. mono.	0.20	26.74	19.41	32.08
DA v2 [Yang et al., 2024]	mono. found.	0.03	27.10	20.20	32.08
DA 3 [Lin et al., 2025]	mono. found.	0.03	27.37	20.90	32.08
DepthPro [Bochkovskii et al., 2024]	mono. found.	0.01	28.40	23.35	32.08

trained-model row), but this is an evaluation-set artefact rather than a training failure. The pseudo-label mIoU is computed against the 27-class CAUSE-TR vocabulary used for label generation; the trained-model mIoU is computed under the standard 19-class Cityscapes evaluation, where stuff classes outside the 19-class set are remapped to void. The change in evaluation cardinality depresses mIoU even as PQ rises, so we treat cross-row mIoU as auxiliary and do not use it as evidence for the main claim.

4.3 Ablations

Three ablations isolate the Stage-1 design.

Component contribution. Table 2 reports DCFA $\times$ SIMCF-ABC at the fixed training-data instance configuration ($\tau_d = 0.20$, $A_{\min} = 1000$, three iterations of dilation). All four cells share the same depth instances; only the pre-clustering adapter and the post-instance filter are toggled.

DCFA and SIMCF-ABC are not interchangeable: DCFA only contributes +0.85 PQ_{th} alone, SIMCF-ABC only contributes +1.33 PQ_{th} alone, and combining them widens the gain to +2.39 PQ_{th} — a margin larger than either alone. The combination is consistent with the two stages targeting different failure modes named in Section 3: DCFA reshapes the partition before clustering by injecting depth-conditioned features into the frozen semantic code, while SIMCF-ABC removes pseudo-instances on which semantic, instance, and depth cues disagree after label generation. The +2.39 PQ_{th} gain concentrates on the things split, where cross-modal disagreement does the most damage and where the filter has the most to fix.

Depth model. Table 3 sweeps the monocular depth source at the standalone-optimum threshold per estimator (semantics held fixed at the $K = 80$ baseline, no DCFA, no SIMCF-ABC). Every depth source improves

PQ_{th} over the no-depth connected-component baseline.

A reader comparing Tables 2 and 3 will notice DepthPro at $\tau = 0.01$ reaches 28.40 PQ in Table 3, but the DCFA + SIMCF-ABC row at $\tau = 0.20$ in Table 2 only reaches 25.85 PQ. The difference reflects a constraint that detector-side training uncovered: at $\tau = 0.01$, average instance count rises to roughly 57 per image with more than half smaller than 1000 px; the resulting fragmented instance set degrades Cascade Mask R-CNN training because the detector overfits to noisy small-instance proposals. At $\tau = 0.20$, count drops to roughly 22 large instances per image and downstream training PQ improves. Table 3 therefore reports each estimator’s standalone optimum, while the rest of the paper uses the training-data configuration; both are correct, in different regimes.

To verify that the full pipeline — not just the bare gradient threshold — generalises across estimators, we re-ran DCFA + SIMCF-ABC end to end with Depth Anything 3 substituted for DepthPro at the same training-data configuration. The DA3 path reaches 26.44 PQ on Cityscapes val (PQ_{th} 18.37, PQ_{st} 32.31, mIoU 55.30), against DepthPro’s 25.85 PQ at the same config. The two estimators trade off in expected directions — DA3’s sharper boundaries lift PQ_{th} by +3.67, DepthPro’s smoother depth gives slightly better stuff coherence — but both reach the same overall regime, which is the multi-estimator generalisation evidence the standalone Table 3 cannot supply.

Encoding sanity. A complementary check (reported in full in the appendix) compares depth-encoding choices on the $K = 300$ semantic clustering protocol. Sinusoidal encoding wins at +2.35 mIoU over no-depth concat, +0.94 over raw concat, and +1.27 over Sobel concat. This justifies the Section 3.3 design choice of injecting depth as a small sinusoidal residual rather than a raw scalar or a hand-engineered gradient.

Table 4: Driving-domain zero-shot transfer (no fine-tuning).

Dataset	Class relation	PQ	PQ _{th}	PQ _{st}	mIoU
Cityscapes (source)	source	35.83	36.26	35.56	44.56
KITTI [Geiger et al., 2012]	aligned	34.85	31.94	36.40	46.87
Mapillary Vistas v2	aligned	39.19	32.06	44.37	58.87

Table 5: Sanity/degenerate-class-space rows; PQ values NOT directly comparable to Table 4.

Dataset	Why separated	PQ	PQ _{th}	PQ _{st}	mIoU
MOTSCheck [Voigtlaender et al., 2019]	2-class subset	61.10	25.51	96.68	92.19
COCO-Stuff-27 [Caesar et al., 2018]	disjoint classes	7.83	7.83	7.84	14.22

4.4 Generalization and Failure Analysis

We split the cross-dataset evaluation into two regimes because they answer different questions. Table 4 reports zero-shot transfer to driving-style targets whose class spaces genuinely overlap Cityscapes; Table 5 is a sanity / degenerate-class-space group whose PQ values are not directly comparable to Table 4.

KITTI and Mapillary Vistas v2 show that the trained detector transfers across the driving class space without retraining: KITTI matches Cityscapes within 1 PQ, and Mapillary Vistas v2 actually exceeds Cityscapes at 39.19 PQ, driven by a 44.37 PQ_{st} on Mapillary’s larger and more varied stuff vocabulary. The MOTS row is reported for transparency but should not be read as a transfer score: MOTS evaluates only two thing classes (pedestrian, car), the dominant stuff classes match Cityscapes’ driving stuff vocabulary, and PQ_{st} = 96.68 inflates the overall PQ to 61.10 — the number reflects in-domain quality on a near-trivial class subset, not an across-the-board improvement. The COCO-Stuff-27 row is the point of this design choice: a Cityscapes-derived $K = 80$ pseudo-label vocabulary cannot recover categories the supervision never represented, and the resulting 7.83 PQ marks the limit of a fixed pseudo-label vocabulary. This is a fixed-class supervision limit, not a generalisation failure — extending to non-driving domains requires a re-labeling step or an open-vocabulary semantic head.

Per-class failure analysis. Table 6 localizes residual errors with Cityscapes val per-class PQ for the final Stage-3 model. The pattern splits into three groups.

Three groups emerge. Large stuff (road 92.99, sky 86.05, vegetation 84.70, building 83.54) and large vehicles (car 70.71, bus 76.67, train 77.17) are strong, because frozen semantic codes give coherent regions and depth boundaries roughly coincide with object extent. Thin structures (pole 2.05, traffic light 6.20) and recognition-bottleneck classes (person 13.37, rider 22.94) are weak: monocular depth provides no separating discontinuity for co-planar pedestrians, and thin structures occupy too few patches to register reliably in the 90-D semantic code.

Seven classes are effectively dead — parking, guard rail, tunnel, polegroup, caravan, trailer, and motorcycle (PQ 0.10) — because the $K = 80$ pseudo-label vocabulary after Hungarian alignment to the 27-class evaluation contains few or no positive examples for them, so the trained model never receives supervision on these categories. The motorcycle case is the cleanest illustration: SQ = 100.00 on the rare match indicates correct mask quality where a prediction lands, but RQ collapses to 0.10 because instances are mass-missed at recognition time.

Two qualitative metrics from the SIMCF-B feature-matching stage close the analysis. Average pseudo-instance count per image drops from 44 before SIMCF-B to 22 after, median instance size grows from 5,502 to 14,965 pixels, and stuff contamination falls from 50.7% to 28.0%. These numbers explain why feature-guided merging improves thing PQ: many depth gradients are intra-object surface changes rather than object boundaries, and merging fragments whose semantic identity and DINOv3 appearance agree recovers the single-object structure that monocular depth had over-fragmented. The dead and weak classes in Table 6 mark the regime where this structural recovery cannot help: when supervision is absent or the category occupies too few patches, no amount of cross-modal agreement repairs it from a single frame.

5 Discussion

Three observations frame the result. First, agreement is the operation that earns the gap. The +3.07 PQ contribution-isolated gap on Cityscapes (35.83 vs 32.76 under the DINOv3 ViT-B/16 backbone) comes from one component: SIMCF-ABC, whose role is to require independent monocular cues to agree. The same is true of DCFA: it does not produce new categories; it bends the frozen partition toward depth-consistent regions where appearance fails. Neither component is a new architecture, and neither component depends on cross-modal training. Both are designed to expose agreement; CUPS-style bootstrapping then amplifies whatever the filter preserves.

Second, the monocular regime is sufficient for scene-centric panoptic segmentation. CUPS commits its pseudo-label generator to stereo depth and unsupervised scene flow. We replace those cues with a single monocular depth map and a frozen self-supervised feature extractor, recover comparable Stage-1 panoptic quality, and exceed the published CUPS detector PQ under a stronger backbone. The same-backbone control belongs to Section 4, but the principle is established: stereo + motion is not load-bearing for the cues CUPS extracted from it.

Third, the gap to supervised performance is dominated by thing-class boundaries. Even with the full pipeline, Stage-1 PQ_{th} on Cityscapes is 14.70, and Stage-3 PQ_{th} remains well below supervised state of the art.

Table 6: Per-class PQ breakdown on Cityscapes val for the final Stage-3 model.

Class	Type	PQ	Class	Type	PQ	Class	Type	PQ
road	stuff	92.99	car	thing	70.71	terrain	stuff	35.68
sky	stuff	86.05	bus	thing	76.67	traffic sign	thing	37.19
vegetation	stuff	84.70	train	thing	77.17	wall	stuff	32.32
building	stuff	83.54	truck	thing	62.64	fence	stuff	20.26
sidewalk	stuff	62.44	bicycle	thing	38.99	bridge	stuff	17.21
			rider	thing	22.94	rail track	stuff	8.40
			person	thing	13.37	traffic light	thing	6.20
			motorcycle	thing	0.10	pole	thing	2.05
			parking	stuff	0.00	guard rail	stuff	0.00
			tunnel	stuff	0.00	polegroup	stuff	0.00
			caravan	thing	0.00	trailer	thing	0.00

Person and rider remain the hardest classes, both because monocular depth produces co-planar fragmentations on closely-spaced pedestrians and because CAUSE-TR’s appearance-driven semantic code fails to distinguish individual persons from a crowd. The DINOv3 backbone change at Stage-2 closes part of this gap, but a fundamentally finer instance signal — single-image depth that resolves co-planar separations, or a learned instance prior conditioned on individual-person geometry — would be required to close the rest.

6 Limitations

Single-seed evaluation. The 35.83 PQ headline reports a single training run from a single seed. Standard error bars across three seeds would strengthen the claim, and a same-backbone CUPS rerun (CUPS labels + DINOv3 ViT-B/16 detector under the inherited recipe) is the most decisive controlled comparison; it is currently pending under compute constraints.

Single-dataset training. Our pseudo-label generator and Stage-2 detector are both fit on Cityscapes, and the cross-dataset transfer experiments in Section 4 are zero-shot. A train-on-source / fine-tune-on-target protocol would isolate which components transfer cleanly and which require domain-specific re-fitting.

Depth-network dependency. SIMCF-C’s depth-statistics rejection and the Section 3.2 instance branch both assume that the monocular depth network produces locally Lipschitz depths inside object interiors. DepthPro [Bochkovskii et al., 2024] satisfies this assumption empirically; weaker monocular networks would degrade SIMCF-ABC’s gating decisions. Our depth-model ablation in Section 4 reports the magnitude of this dependency for DA3, SPIdepth, and ZoeDepth.

27-class CAUSE + Hungarian evaluation protocol. Following CUPS, we evaluate against the 27-class CAUSE pseudo-vocabulary with a Hungarian map to Cityscapes’ 19 classes. Cross-paper comparison with works that report standard 19-class panoptic quality requires reading

the protocol carefully; the comparison numbers reported in Section 4 are on the matched protocol.

7 Broader Impact

Unsupervised panoptic segmentation reduces the labelling-cost barrier for dense scene understanding, and the proposed monocular pseudo-label pipeline removes the additional sensor barrier that stereo and unsupervised scene flow impose. The immediate beneficiaries are autonomous driving systems for which calibrated stereo or paired video is unavailable, and robotic perception platforms that operate from a single monocular camera. Accessibility tools that rely on dense scene parsing — wayfinding, hazard alerting, robotic assistance — also stand to benefit from a recipe that no longer needs multi-view capture.

Two risks deserve naming. Surveillance applications. Lowering the labelling-cost barrier for dense scene understanding also lowers the cost of large-scale automated surveillance. The same recipe that supports privacy-preserving analytics can support intrusive monitoring; deployment in privacy-sensitive settings should follow human review and explicit consent.

Dataset-bias amplification. Pseudo-labels inherit the biases of the frozen priors that produce them, and those priors carry forward biases from their pretraining corpora. A monocular pipeline that depends on DINOv2 / DINOv3 and DepthPro carries forward whatever class-distribution and demographic biases those models encode. Safety-critical deployment (autonomous driving, medical imaging) should validate against ground-truth annotations stratified by class and by demographic subgroup before downstream use.